ViewPoint: Panoramic Video Generation with Pretrained Diffusion Models

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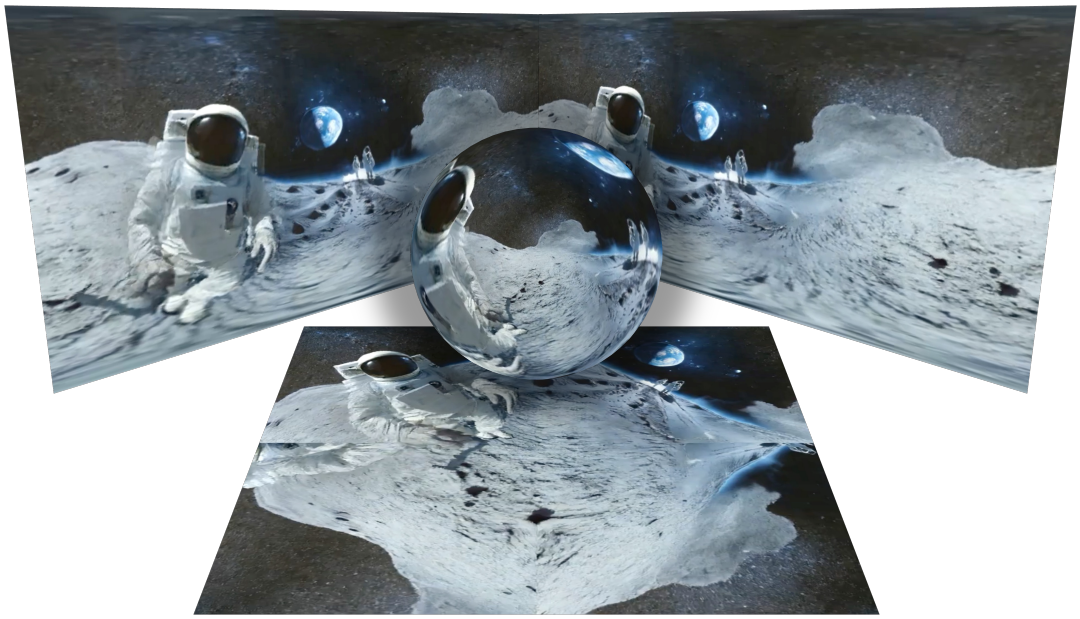


Figure 1: The generated result. The image at the bottom displays the ViewPoint map generated by our model, with the background image showing the concatenated equirectangular map derived from the ViewPoint map. The generated panoramic image exhibits excellent spatial consistency, as the equirectangular map can be seamlessly stitched together. Project page: ViewPoint.

Abstract

Panoramic video generation aims to synthesize 360-degree immersive videos, holding significant importance in the fields of VR, world models, and spatial intelligence. Existing works fail to synthesize high-quality panoramic videos due to the inherent modality gap between panoramic data and perspective data, which constitutes the majority of the training data for modern diffusion models. In this paper, we propose a novel framework utilizing pretrained perspective video models for generating panoramic videos. Specifically, we design a novel panorama representation named ViewPoint map, which possesses global spatial continuity and fine-grained visual details simultaneously. With our proposed Pano-Perspective attention mechanism, the model benefits from pretrained perspective priors and captures the panoramic spatial correlations of the ViewPoint map effectively. Extensive experiments demonstrate that our method can synthesize highly dynamic and spatially consistent panoramic videos, achieving state-of-the-art performance and outperforming previous methods.

Introduction

Imagine you're traveling—would you seize the chance to capture the breathtaking landscapes that unfold before you? When you return home, do you long to relive and immerse yourself

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in those vivid experiences once more? Recently, omnidirectional vision has garnered increasing attention as it unlocks immersive AR/VR, virtual travel, and telepresence experiences. However, recording 360-degree videos requires expensive professional devices, i.e., 360 cameras, making the creation of panoramic content challenging, and most consumers capture only narrow field-of-view (FoV) clips on portable monocular cameras, e.g., smartphones. Enabling these perspective recordings to become full panoramas would democratize spherical media, letting anyone relive or share memories in true 360-degree form.

Achieving this conversion is non-trivial due to fundamental representation gaps between perspective and panoramic domains. A straightforward approach is to adopt the widely-used Equirectangular Projection (ERP) for panoramas, which maps the spherical view onto a rectangular image. Unfortunately, this format introduces severe distortions, especially near the poles, stretching and squashing content unnaturally. More critically, equirectangular images lie outside the distribution of visual training data for modern generative models. Diffusion models [10, 21, 24] and VidLs [14] are predominantly trained on perspective imagery, so without significant adaptation they struggle to produce high-quality results in the warped ERP space. On the other hand, one could represent the panorama as multiple perspective projections. A common choice is the Cubemap Projection (CP) format, which unfolds the panoramic sphere onto six faces of a cube (each 90° perspective view). This representation avoids polar distortions and yields locally planar patches well-aligned with the priors of convolutional and diffusion networks. However, a naive cubemap suffers from spatial discontinuities at the borders of adjacent cube faces. This is because the design in the CP is a fundamental model to capture cross-face consistency. In short, existing representations force a trade-off between continuity and distortion: ERP offers continuous but distorts the content, whereas CP preserves local fidelity but fragments the panoramic space. Neither is ideal for generative video modeling, especially when temporal consistency is also required.

In this paper, we address the above mentioned issues by introducing ViewPoint, a novel 360° video representation and generation framework that bridges the strengths of ERP and CP. At the core of ViewPoint is a specially-encode panoramic map that reflects the spherical structure. This capable of stitching into an overlapping views with greatly improved continuity. Specifically, we first convert the panorama into six cube faces and a distortion-free perspective views and then merge them into a small number of overlapping "pseudo-perspective" panels. Because these panels share content along their boundaries, they eliminate the hard seams of a standard cubemap while remaining fully planar. Building on this representation, we propose a Pano-Perspective attention that allows a two-key benefits: (1) Global coherence: Pano-attention spans the entire stitched map (and time), ensuring that opposite directions align and the whole panorama flows in a plausible way; (2) Local fidelity: within each panel, Perspective-attention blocks focus on individual patches to preserve details finely, matching the strong image priors that our large panoramic model provides.

In summary, our contributions are as follows:

- We introduce a novel representation for 360 degree content aimed at improving spatial continuity and reducing distortion while leveraging the power of diffusion models to generate 360-degree videos through the proposed format.
- We design a Pano-Perspective attention mechanism, enabling the model to simultaneously maintain global spatial continuity across the entire panorama and significantly enhance the preservation of fine-grained details and motion.
- Extensive experiments demonstrate that our method can generate high-quality 360-degree videos and achieve state-of-the-art performance, outperforming previous approaches.

Related Works

Panorama Representations. Panoramic images are often projected from spherical space to the 2D plane for processing and storage [1]. Equirectangular Projection (ERP) is the most popular format, which uniformly maps pixels from a spherical surface to a planar rectangle. Despite its simplicity and spatial continuity, ERP inevitably brings disadvantages such as geometric distortion at the poles. Cubemap Projection (CP), on the other hand, projects panoramas to six cube faces with different FoV of 90° to alleviate geometric distortion. However,

under this representation, only one cube face is explicitly spatially continuous with its four adjacent faces on the 2D plane.

Panorama Generation. Panoramic image generation [3, 26, 32, 33, 8, 6, 18, 34, 26, 22, 29, 35] aims to synthesize 360-degree immersive images based on user-provided textual or visual clues. It generally requires panoramic images as data, while the panoramic images are usually captured in an ERP format, most methods perform diffusion process on the ERP format. Recent works [26, 22, 20, 32, 34, 30, 29, 13] leverage powerful diffusion models to generate high-quality panoramic images. Panfusion [34] proposes a dual-branch architecture to generate panoramas in both perspective and ERP format. Recently [8, 26, 33, 20, 26] introduces LoRAs [15] achieves high-quality results in generating panoramic images. However, a naive cubemap suffers from spatial inconsistency across the boundaries of adjacent perspective projections. Cube2Diff [13] uses an alternative representation, cubemap, to synthesize panoramic images with high fidelity and diversity. Despite the success of significant progress in omnidirectional image generation, achieving similar success in the field of video remains a challenging problem, as it requires spatial-temporal consistency across the entire spherical space, as well as costly computational resources.

A few works [31, 16, 19, 25] explore panoramic video generation. 4K4DGEN [16] animates a given high-resolution panoramic image through user intention. However, the requirement of high-quality panoramic image cannot be applied to a variety of scenes. 360DVD [31] extends a given video to a panorama through 360-adaptor with LoRA. But the trained on ERP-formatted content without the sufficient spatial continuity, and the content of the results limited by the LoRA priors. Imagine360 [25] trains a 360-Adaptor to exploit the generative capabilities of text-to-video generation models trained on ERP-formatted panoramas that suffer from the spatial discontinuity. Unlike previous work, our method reasonably explores the boundary of pretrained perspective video generation models, which is inspired by a novel representation panoramas with a novel perspective video generation model, which is inspired by the strong image priors that our large panoramic model provides.

Video Outpainting. Formally, extending from a given perspective view to a panoramic video is a kind of video outpainting task. Benefiting from pretrained diffusion models [10, 21, 22, 9], previous works [28, 5] have made notable progress in perspective video outpainting. However, due to the challenging equirectangular format. Following the paradigm of previous works [28, 5], we propose a novel panoramic video outpainting method within a diffusion process at the boundaries of the sub panoramic video generation.

Method

Preliminary

Latent diffusion models [23, 10, 21] conduct a series of diffusion and denoising processes in latent space. Given a clean latent code x_0 from the training data, a noisy latent x_t is obtained by adding an random noise $\varepsilon \sim \mathcal{N}(0, I)$ to x_0 according to a timestep $t \in [0, 1]$. Following the flow matching paradigm [17, 7], x_t is defined as

$$x_t = tx_1 + (1-t)x_0$$

The training objective of the model is to predict the velocity v_t, thus, the loss function of the training process can be formulated as

$$L = \mathbb{E}_{x_0, x_1, t, c_{\text{txt}}, c_{\text{vis}}} \parallel v_\theta(x_t, t, c_{\text{vis}}, c_{\text{txt}}) - v_t \parallel^2$$

where c_{vis} is the text condition, θ is the model weights, u is the predicted velocity and v_t is the ground truth velocity.

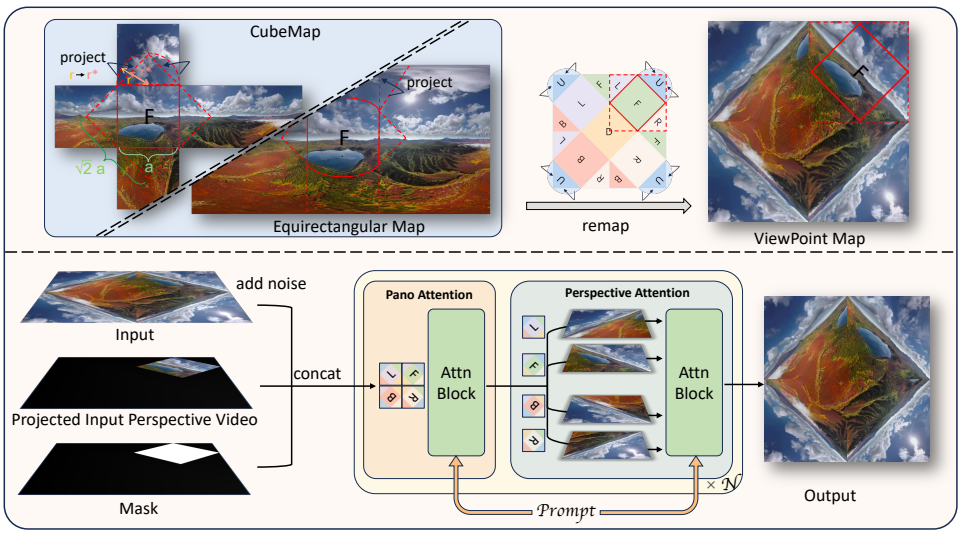


Figure 1: **Method overview.** The first row illustrates how to construct a Viewpoint map from a CubeMap or an Equirectangular Map. We begin by constructing subregions, using F as an example, we construct a pseudo perspective region called V. Subsequently, we combine the four subregions to form a ViewPoint map. The second row shows the pipeline design: we first concatenate the noisy Viewpoint map, input video, and relevant mask along the channels dimension, and then employ a Pano-Perspective attention mechanism to learn how to generate spatial-temporal consistency while modeling fine-grained visual information.

ViewPoint Map

To maintain spatial continuity while reducing spatial distortion, we propose ViewPoint Map, a novel panorama representation that combines the advantages of ERP and CP. We first sample six cube faces from either an equirectangular map or a cubemap, namely, F, R, B, L, U, and D, which represent the front, right, back, left, up, and down faces, respectively. The four faces in the horizontal view, i.e., F, R, B, L, typically contain most of the visual content that users focus on. Therefore, we construct pseudo-perspective subregions centered around these four faces. As shown in the first row of Fig. 2, we select one face as the central face (assuming the model provides), and finally, the central face and its four adjacent parts are concatenated to form a square region with a side length of $\sqrt{2}a$. This design makes the central face more spatially continuous with the four adjacent parts. For example, if F is the central face, then its four adjacent faces are L, U, R and D, located to the left, right, above, and below F respectively. After obtaining the four subregions, we perform stitching to form the ViewPoint Map. As illustrated in the second row of the ViewPoint Map (the M face) faces also continues. To address the splitting of U, we project their adjacent to the central face are adjacent to the standard mapping data. After all, generating perspective videos with extreme motion is still a challenge, let alone panoramas.

then the scaled radial coordinate r^* is computed by

$$d(\theta) = \frac{4r}{\sin\theta + \cos\theta}$$

$$r^* = \frac{r \cdot d(\theta)}{a}$$

This design allows the U faces of the four subregions to have a small overlapping portion, thereby ensuring internal consistency of the entire U face.

Pano-Perspective Attention

ViewPoint Map offers global panorama information and effectively utilizes the inherent in-context generation capabilities [11] of diffusion models. To further accelerate convergence and

exploit the priors of the base model, we propose a Pano-Perspective Attention mechanism. Specifically, Pano-Attention is responsible for contextual learning to maintain spatial consistency, while Perspective-Attention focuses on generating fine-grained visual information for each subregion. As shown in the second row of Fig. 2, we first concatenate the noisy Viewpoint Map, the projected input perspective view frames, and the corresponding masks along the channel dimension. In the latent space, the shape of the concatenated features is (batch_size, channels, frames, height, width). After passing through the Pano-Attention block, we reshape it to (batch_size, subregions, channels, frames, height_sub, width_sub) as input for Perspective-Attention, so that the model can achieve two functions: (1) generate spatial-temporal consistent panoramic videos by Pano-Attention; (2) achieve high-quality panoramic content within each subregion via Perspective-Attention.

Overlapping Fusion

Each subregion in a ViewPoint map has a partial spatial overlap with the two adjacent subregions. To achieve smoother transitions between subregions, we propose a gradient fusion mechanism. As shown in Fig. 3, the overlapping parts between two subregions form a regular rhombus; therefore we construct an overlapping fusion weight $W \in \mathbb{R}^{h \times w}$ where the values decay from the area near the central face to the edges.



Figure 2: **Overlapping fusion.** The four subregions partially overlap with each other, thus we propose a gradient fusion mechanism to interpolate the overlapping areas, thereby enhancing spatial consistency.

For each subregion $S_k \in \mathbb{R}^{s \times s}$, $s = 2a$, $d \in [F, R, B, D]$, the overlapping fusion process can be described by the following formulas:

$$W_{i,j} = \frac{i + j - s}{2s - 2i - 2j} \quad i, j = 1, 2, ..., s$$

where i, j denote the row and column indices, respectively. Next, we define notations: For a given matrix A, we have

$$R_{\text{lr}}(A) = A^T$$
$$R_{-90}(A) = A \cdot J$$
$$R_{90}(A) = J \cdot A^T$$

where J is the exchange matrix. Finally, for each subregion, we have

$$S_{F_{i,j}} = \begin{cases} S_{F_{i,j}} - R_{90}(W) + R_{-90}\big(S_{R,i,s-j}\big) \cdot R_{-90}(W), & (i,j) \in [1,s] \times [s+1,s] \\ S_{F_{i,j}} - R_{-90}(W) + R_{90}\big(S_{L,i,s-j}\big) \cdot R_{90}(W), & (i,j) \in [1,s] \times [1,s] \end{cases}$$

$$S_{R_{i,j}} = \begin{cases} S_{R_{i,j}} - R_{90}(W) + R_{-90}\big(S_{B,i,s-j}\big) \cdot R_{-90}(W), & (i,j) \in [1,s] \times [s+1,s] \\ S_{R_{i,j}} - R_{-90}(W) + R_{90}\big(S_{F,i,s-j}\big) \cdot R_{90}(W), & (i,j) \in [1,s] \times [1,s] \end{cases}$$

$$S_{B_{i,j}} = \begin{cases} S_{B_{i,j}} - R_{90}(W) + R_{-90}\big(S_{L,i,s-j}\big) \cdot R_{-90}(W), & (i,j) \in [1,s] \times [s+1,s] \\ S_{B_{i,j}} - R_{-90}(W) + R_{90}\big(S_{R,i,s-j}\big) \cdot R_{90}(W), & (i,j) \in [1,s] \times [1,s] \end{cases}$$

$$S_{L_{i,j}} = \begin{cases} S_{L_{i,j}} - R_{-90}(W) + R_{90}\big(S_{F,i,s-j}\big) \cdot R_{90}(W), & (i,j) \in [1,s] \times [s+1,s] \\ S_{L_{i,j}} - R_{90}(W) + R_{-90}\big(S_{B,i,s-j}\big) \cdot R_{-90}(W), & (i,j) \in [1,s] \times [1,s] \end{cases}$$

Note that we apply overlapping fusion to each subregion simultaneously, rather than in the order specified in the above formulas.

For the D face, we first project the triangles back into semicircles and transform the rhombus into a petal shape formed by the overlap of two circles. Finally, we use a similar fusion mechanism to fuse the overlapping of the four semicircles.

Experiments

Implementation Details

Datasets & Preprowess. Our model is trained on 4 panorama datasets, including one image dataset, Flickr360 [3], and three video datasets, WEB360 [31], ODVista [3], and 360+x [4]. Among them, only WEB360 [31] contains text captions; therefore, we use Qwen-VL [2] to annotate the other three datasets, generating corresponding descriptive captions. All datasets are resized to a resolution of 512×1024, and during training, video data is divided into clips of 49 frames each.

Training. We first inflate the patch embedding layer of the powerful video generation model, Wan2.1 [27], from 16 to 33 channels to accommodate the input data, and then fine-tune the entire model. The training process is executed on $8 \times$ NVIDIA A100 GPUs, using a batch size of 1 and a learning rate of 1e-4. We employ a joint image-video training strategy, treating images as videos with only one frame.

Qualitative Comparison

We compare our approach with three previous methods, including one perspective video outpainting method, Follow-Your-Canvas [5], and two 360-degree video generation methods, 360DVD [31] and Imagine360 [25]. Adopting equirectangular representation, both 360DVD [31] and Imagine360 [25] suffer from severe distortion at the poles. As shown in Fig. 4, they appear to have "black holes", while the ground shows "swirls". We argue that these artifacts are caused by the model's difficulty in handling the spatial-temporal distortion introduced by ERP; as a slight movement can result in significant tress at the pole.

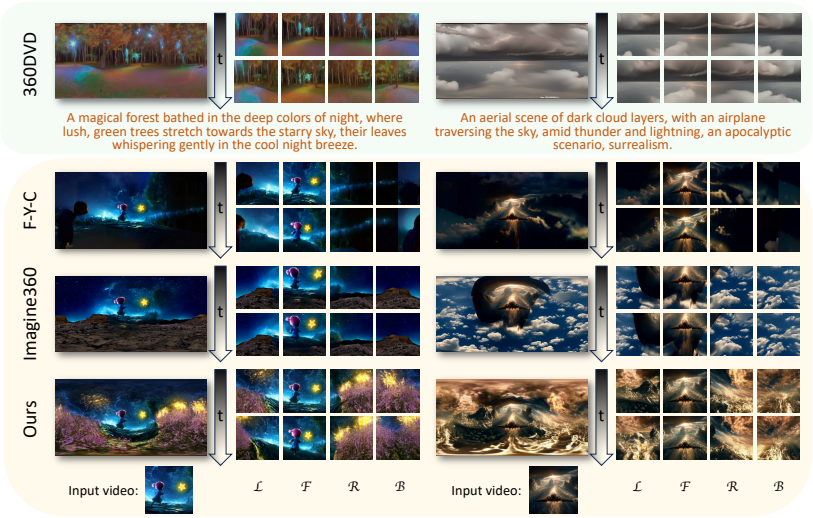


Figure 3: Results from 360DVD [31] and Imagine360 [25]. Despite numerous efforts to mitigate geometric distortion at the poles, both methods still struggle with generating realistic top and bottom views. The distortion at the poles is even more pronounced in video scenes, severely affecting realism and immersion.

The visualization of the comparison results is shown in Fig. 5, where we present the ERP format and perspective views in four horizontal directions, with arrows indicating the flow of time. As shown in Fig. 5, 360DVD [31] only recognizes textual input. The generated frames exhibit terrible image quality and possess little motion. Follow-Your-Canvas© V-C [5] generates satisfactory perspective videos, however, its results suffer from severe spatial inconsistency, especially on the R face, exhibiting a noticeable sense of disconnection. Although Imagine360 [25] represents the coherence of scenes with lower distortion in panoramic space, is highly compromised. Furthermore, it does not appear the ground does not move periodically and shows a noticeable texture difference from the input video. In the second example, the aircraft in the input video is in a dark and mysterious environment, while the generated panoramic video shows blue skies and white clouds, contrary to the given input. Additionally, the aircraft does not integrate well with the surrounding environment, resulting in a discordant effect. In contrast, our approach demonstrates excellent spatial-temporal consistency and supports both textual prompts and input videos, thus enhancing the flexibility of generation. Furthermore, our ViewPoint exhibits richer dynamics; both the water flow and clouds respond to the motion trends of the input video correctly. As in the first case, the input video depicts a little girl running in a wilderness, forward, with the model producing a corresponding effect. Our generated results align with both the input and the prompt, delivering a magical forest scene at night. Our approach exhibits high-quality dynamics, and Imagine360 [25], demonstrating our method's ability to generate satisfactory panoramic content.



Figure 4: Qualitative comparison of generated videos. 360DVD is a text-driven approach and exhibits terrible image quality while Follow-Your-Canvas fails to generate panoramic videos with a reasonable spatial layout. Imagine360 suffers from spatial-temporal discontinuity. Our approach, on the other hand, can generate high-quality panoramic videos aligned with the given input. Due to page limits, we highly recommend watching the dynamic videos available at ViewPoint.

Quantitative Comparison

In this section, we provide a quantitative comparison of our approach with previous methods. VBench [12] is a comprehensive benchmark suite for video generative models which scores a video in 16 dimensions. We evaluate our approach and previous methods on the ODV360 [3] dataset across five dimensions: "subject consistency", "imaging quality", "motion smoothness", "aesthetic quality", "dynamic degree". Specifically, we first project the generated videos into six perspective views with a FoV of 90°, similar to F, R, B, L, U, and D. Then, we apply VBench [12] evaluation in a perspective manner for a comprehensive evaluation.

The results of the quantitative evaluation are shown in Tab. 1, where our approach achieves the best scores across four metrics. Although 360DVD [31] performs the second best in "subject consistency", it scores the lowest in "dynamic degree", indicating that the videos generated by 360DVD have very little motion. Follow-Your-Canvas [5] achieves the highest score on "motion smoothness" but scores poorly on the projected videos evaluated by VBench [12]. This shows that although this method possesses strong dynamic ability, it lacks spatial consistency in panoramic space. Compared with Imagine360 [25], our approach shows superior performance in all metrics, further demonstrating the effectiveness of our method, and we believe ViewPoint can provide valuable insights for the omnidirectional vision community.

Method	subject consistency	imaging quality	motion smoothness	aesthetic quality	dynamic degree
Follow-Your-Canvas [5]	0.8384	0.4464	0.9615	0.5501	0.9500
360DVD [31]	0.8625	0.4634	0.9524	0.5503	0.5083
Imagine360 [25]	0.8547	0.5039	0.9720	0.5348	0.6146
View-Point(Ours)	**0.8757**	**0.5257**	**0.9800**	**0.5601**	**0.9583**

Table 1: Quantitative comparison on Vbench. Our method achieves the best performance in all four metrics. Although 360DVD obtains the second highest "subject consistency" score due to the nearly static nature of its generated videos, the results fail to meet the demands of generating high-quality panoramic videos. In contrast, our approach performs well in both motion consistency and dynamics.

Ablation Study

To demonstrate the effectiveness of our method, we conduct ablation experiments on panorama representation formats and network designs. For the ERP format, we directly fine-tune the base model to adapt to this panoramic representation. For the CP format, we train two models, CP-noloRA and CP-4DRoPE, separately to assess the impact of different cubeface-encodings. Technically, CP-loRA rearranges the six cube faces into a 2-row by 3-column rectangle in raster order, i.e., F, R, B on the first row, and D, L, U on the second row, while CP-4DRoPE assigns an independent 3D positional encoding to each cube face. As for the network designs, we compare our design of Pano-Perspective attention with two variants: "imaging quality", "motion smoothness", and "aesthetic quality". As shown in Fig. 3, the overlapping parts are eliminated (w/o Pano-Attn); or we eliminate the perspective-attention (w/o Pers-Attn). Additionally, we compare quantitative results.

The results of quantitative ablation are shown in Tab. 2, our full method outperforms the other alternative designs across four metrics. Since we conduct evaluations in a perspective manner, ERP-3D-LoRA performs best in "imaging quality". However, as shown in Fig. 6, though such perspective videos show good visual quality, there is still a discontinuity issue at the boundaries of the four faces. Therefore, it does not meet the broad usage of the high-quality panoramic video generation.

	subject consistency	imaging quality	motion smoothness	aesthetic quality	dynamic degree
ERP	0.8712	0.5195	0.9781	0.5504	0.8917
CP-4DRoPE	0.8621	0.5104	0.9736	0.5433	0.8730
CP-ICLoRA	0.8727	**0.5340**	0.9770	0.5525	0.8792
w/o Pano-Attn	0.8664	0.5122	0.9713	0.5410	0.9028
w/o Pers-Attn	0.8631	0.5109	0.9727	0.5464	0.9042
Full Method	**0.8757**	0.5257	**0.9800**	**0.5601**	**0.9583**

Table 2: Quantitative ablation. Compared to the other four alternative designs, our approach achieves the highest scores across three metrics. Although CP-ICLoRA achieves the highest score in "imaging quality", proving that each of the six faces has good quality independently, the panoramic composition of these six faces exhibits noticeable discontinuities, as indicated by the red boxes in Fig. 6. Therefore, it does not meet the demand for high-quality panoramic video generation.

User Study

Despite achieving advanced scores on Vbench [12], we conduct user studies, introducing subjective user ratings to further validate the superiority of our approach. We ask participants to vote on the tested videos based on four dimensions: "Spatial Continuity", "Temporal Quality", "Aesthetic Preference", and "Condition Alignment". "Spatial continuity" represents the coherence of scenes and structures in panoramic space, including their smoothness in 360°. "Aesthetic Preference" represents the projected videos evaluated by VBench [12]. This shows that although this method possesses strong dynamic ability, it lacks spatial consistency in panoramic space. "Condition Alignment" refers to the degree to which the generated panoramic video aligns with the input conditions, such as text or video.



Figure 5: Ablation on different designs. ERP exhibits serious artifacts due to the natural gap in modality. Both Cube representation methods have spatial discontinuity issues. Without Perspective-Attention, it leads to misalignment with the input video. Our full method can generate reasonable and spatially consistent results.

Finally, we collect 50 valid questionnaires, each containing 14 sets of videos for comparison, and the results are shown in Fig. 7. Overall, Imagine360 [25] is the second choice among participants, while 360DVD [31] and Follow-Your-Canvas [5], receive the fewest votes. Our approach receives the highest number of votes across all four dimensions, surpassing 360DVD [31], Follow-Your-Canvas [5], and Imagine360 [25], demonstrating our method's ability to generate satisfactory panoramic results.

Conclusion

In this work, we present ViewPoint, a novel framework for representing and generating panoramic videos leveraging modern generative diffusion models. Specifically, we design a novel representation distinct from traditional panoramic image and video representations, which has the advantages of both spatial continuity and temporal consistency. Through our proposed Pano-Perspective attention mechanism, our trained model preserves the global spatial-temporal continuity of the generated panoramic videos. With our proposed Pano-Perspective attention mechanism, our trained model perceives the global panoramic space, effectively leveraging the strong priors of pretrained perspective video generation models. Extensive results, both qualitative and quantitative, show that our approach outperforms previous methods. We look forward to the effectiveness of our method, and we believe ViewPoint can and Imagine360 [25], demonstrating our method's ability to generate satisfactory panoramic results.

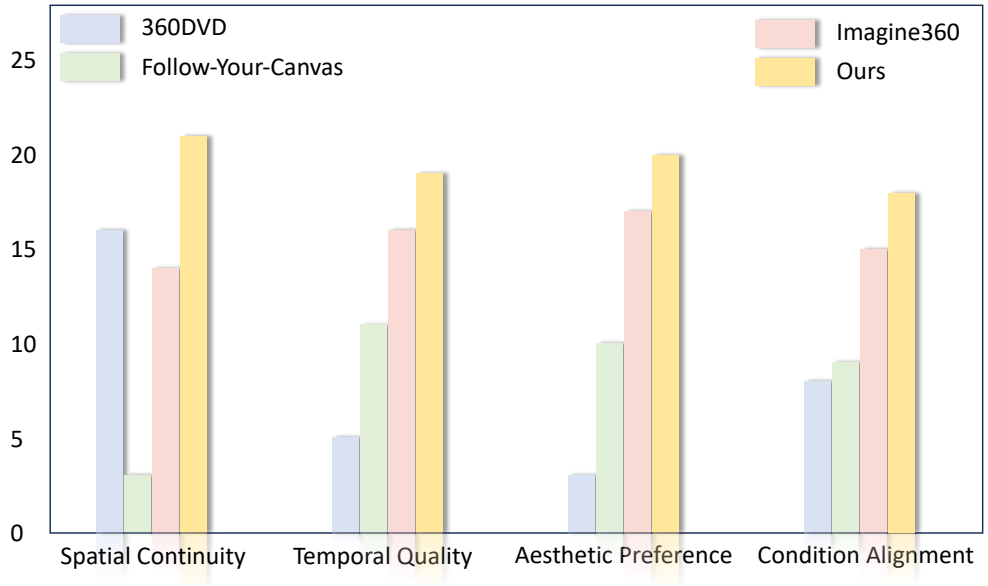


Figure 6: User studies. We ask participants to vote on the videos generated by four methods based on four dimensions, and our approach receives the most votes.

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